

ESIM Wiring Test

Does the ESIM switch signal reach the engine computer intact?

2018 Alfa Romeo Stelvio 2.0T (GU) | VIN ZASFAKPN5J7B88115 | about 142,290 km

Why this test

Three smoke tests (08/2025, 2026-09-16, 2026-09-26) show the EVAP system is sealed, and the canister, ESIM (04861961AD), filter, fuel cap and purge valve are all new. Yet the ECU keeps setting P0455, P0456 and P0440. The ECU decides "leak" when the ESIM switch does not report closed during its test. On a sealed system with a new ESIM, the remaining causes are the ECU's calibration and the **signal path from the ESIM to the ECU**: connector, wires, ground, and the ECU input. This test checks that path.

Do the wiTECH flash check first (ECM Flash tab, Current vs New ECU Part Number; see the 'Stelvio ECM Flash Procedure (wiTECH)' PDF). It takes minutes and decides whether this test is needed.

What is not known here. The factory pinout, wire colours, ECU connector and pin numbers, and the switch's rated voltage for this car were not obtainable. Look them up in the FCA wiring diagram for this VIN (FCA service information via your wiTECH/TechAuthority access) and write them in the table on page 2 before testing. Every value to compare against comes from that diagram or from a known-good reading, never from this sheet. General circuit-test practice below is marked *[practice]*.

One clue from the codes: the ECU has not set an ESIM *circuit* code (for example P24D6, EVAP pressure switch circuit range/performance, which TSB 18-026-20 lists for this car). A hard open or short usually sets a circuit code, so an intermittent or high-resistance fault is more likely than a broken wire.

Tools

- wiTECH 2 with your interface (for the ECU's live reading of the switch).
- Digital multimeter; back-probe pins or a breakout lead. *[practice]* Never push probes into the front of connector terminals; it spreads them.
- Hand vacuum pump and short hose (as for the ESIM bench test).
- A fused jumper lead (low-amp fuse). *[practice]*
- The FCA wiring diagram for this VIN.

Circuit details from the wiring diagram (fill in first)

Item	Value from the FCA diagram for this VIN
ESIM connector ID / location	
Signal pin and wire colour	
Ground pin and wire colour	
ECU connector and signal pin	
Ground point (e.g. G-number)	
Switch reference voltage	
wiTECH parameter name for switch state	

Test A: end-to-end, with the ECU as the meter (best first test)

This proves switch, connector, wires and ECU input in one go, without unplugging anything.

A1. Find the switch state in wiTECH

Ignition RUN, engine off. In wiTECH open the ECM data display and find the parameter that reports the ESIM / EVAP pressure switch state (name varies; look under EVAP or leak detection).

A2. Apply vacuum at the ESIM

Behind the driver-side rear wheel liner, disconnect the hose on the ESIM's canister side and apply gentle vacuum to that port with the hand pump. Leave the ESIM plugged in.

A3. Watch the ECU's reading

- **Switch shows CLOSED with vacuum and OPEN when released, every time:** the whole signal path works. The wiring is exonerated; the fault is the ECU calibration or its test logic. Take it to the flash check / dealer.
- **Never changes:** the ECU is not seeing the switch. Go to Test B to find where it is lost.
- **Changes, but drops out when you wiggle the harness or connector:** intermittent connection. Go to Test B, steps B1 and B5.

Test B: isolate the break

B1. Inspect the connector

- Ignition OFF. Unplug the ESIM connector.
- Look for green or white corrosion, water, bent or pushed-back pins, a cracked housing or a missing seal. *[practice]* Check terminal grip with a matching test terminal: it should drag, not slide.
- Follow the harness back as far as practical: chafe points on the wheel-well edge and subframe, and rodent damage.

B2. Signal wire: is the ECU sending its reference?

Connector unplugged, ignition RUN. Back-probe the harness-side **signal** pin (from the diagram) to a good chassis ground.

- **Expected:** the reference voltage shown in the diagram (the ECU's pull-up on the switch line). Record it.
- **0 V:** open in the signal wire, or the ECU is not driving it: go to B4.
- **Battery voltage where the diagram shows a low reference:** short to power: go to B4.

B3. Ground wire

Ignition RUN, connector unplugged. Measure between the harness-side **ground** pin and the battery negative post.

- *[practice]* A good ground reads close to 0 V and a few ohms or less (ignition off, for the ohms reading). Record it and compare with a known-good ground on the car.

B4. End-to-end resistance of each wire

Ignition OFF and battery negative disconnected before unplugging the ECU connector. Measure each wire from the ESIM connector pin to its ECU connector pin (from the diagram).

- *[practice]* Each wire should read a fraction of an ohm. Several ohms or more is high resistance; open line is a break.
- Also check each wire to chassis ground (should be open) and to battery positive (should be open): a reading means a short.
- Reconnect the ECU connector carefully and the battery last.

B5. Jumper test: does the ECU see a closed switch?

Battery reconnected, ignition RUN, ESIM unplugged, wiTECH data display open. With the fused jumper, connect the harness-side signal pin to the ground pin (this is what the closed switch does).

- **wiTECH shows CLOSED:** wiring and ECU input are good. If Test A failed, the ESIM switch or its connector contacts are at fault.
- **wiTECH still shows OPEN:** the fault is in the wiring or the ECU input. Repeat the jumper at the ECU end (from the diagram) to split wiring from ECU.
- *[practice]* Wiggle the harness along its length while watching: a flicker finds an intermittent fault.

Record sheet

Test	Result
Date / odometer	
A3: wiTECH state with vacuum / released	
A3: wiggle test	
B1: connector condition	
B2: signal reference voltage	
B3: ground voltage / resistance	
B4: signal wire resistance, ends / to ground / to B+	
B4: ground wire resistance	

Test	Result
B5: wiTECH state with jumper	
Conclusion	

After the test

- **Wiring fault found:** repair it (FCA-approved terminal and splice repair), clear codes, drive several days with fuel 15-85 percent and cold starts, then confirm with cuore's verify_repair: all three codes 'passed since clear'.
- **Signal path good (Test A passes):** the fault is in the ECU calibration or test logic. Flash check first; if already current, a dealer case with these three clean smoke tests and this sheet is the evidence.
- Enter the readings in cuore's evidence gate as manual measurements; a Test A pass is the disconfirming test that clears the wiring.